

You Can't Outsource Production Judgement

Why AI Makes Unreliable Narrators of Us All

Joe Woodhouse | DevOps Not Dead | London 2026

What We're Covering Today

Need 90 mins to do this properly.
Could rush it in 60. I have 25.

I'll go deep on what I can.
The rest is in the appendices.
You'll get the complete deck.

Act I

TL;DR

Bottom Line Up Front

LLMs are unreliable narrators

... And they make us one, too.

Individual risks vary with the amount of outsourcing

... Counter-intuitively, mid-level is the riskiest.

Risks cascade: individual to department to company.

Why You Don't Believe Me

We feel faster with genAI LLM.

We believe we're still doing the cognitive work.

So did they.

Why You Don't Believe Me, cont.

METR Study (2025), gold-standard RCT

Experienced devs. Real paid work on real repos.

Subjective self-reporting: 24% faster.

Objective measurement: 19% slower.

They didn't notice a 43% performance gap. Why?

Outsourcing Hierarchy

Level 1: AI Assist, or “Help me as I do it”

Semantic search, syntax completion

Level 2: AI Augment, or “Do some of it for me”

Prompt-by-prompt code gen, debugging, testing

Level 3: AI Outsource, or “Do it, tell me when ready”

Vibe coding, agentic, autonomous workflows

Outsourcing Hierarchy, cont.

Counter-intuitive finding, independent studies:

Level 2 is the most dangerous to individuals.

- We still engaged

- We believe we're still doing the work

- AI substitutes for our skill while we watch

- Skill decay occurs outside our awareness

Outsourcing Hierarchy, cont.

Level 3 risks are catastrophic...
but nobody's fooled about contribution

A Touchstone Test

Can we explain the answer and how it was reached
... without going back to check the LLM chat?

Yes: it's our answer, we learned, we can recall it.

No: We used AI instead of skill.
... and we may have lost something

Act II

“Extraordinary Claims Require
Extraordinary Evidence”

The Mechanisms

M1: Unreliable Narration

(M2: Skill Decay)

(M3: Security/Quality Debt)

(M4: Cognitive Overload Paradox)

(M5: Autonomous Agent Catastrophe)

Unreliable Narration - BLUF

We can't trust LLM outputs.

“Hallucinations”; non-determinism; GIGO

We can't trust our own perceptions of LLM use.

Contaminated planning. Metrics. Risk assessment.

This is most dangerous at Level 2: AI Augment.

We don't notice what's happening.

Unreliable Narration - Evidence

METR (2025): The 45% Perception Divergence

Experienced devs, real OSS repos, RCT

Claimed: 24% faster | Actual: 19% slower

Perception diverged 43% from reality

They still believed they were faster post-study!

Unreliable Narration - Evidence

MIT (2025): Cognitive Debt Study

54 students, 4 months, writing SAT essays

LLM users: 83% couldn't recall own essay

EEG showed alpha/beta waves halved

Neural engagement declined over time

Unreliable Narration - Evidence

Frontiers in Psychology (2025): Substitutive Offloading

Users overestimate knowledge of AI output

LLMs create “illusion of competence”

Memory encoding & retrieval decline over time

... User is unaware

Unreliable Narration - Cascade

Harms to individuals harm the Eng department

Impact of 43% perception gap on estimation

Project estimates driven by self-reports

Faros AI study of 10,000 devs

Everyone claimed LLM-driven speedups

DORA metrics flat

Unreliable Narration - Cascade

Harms to Eng department harm the whole org

- Shipping cadence

- Code quality, code effort

- AI metrics likely disconnected from P&L

We already don't usually measure what matters

- Consider AI effects on PRs, lines of code, etc.

Unreliable Narration - Mitigations

1. The Touchstone Test

Explain the working without checking the chat?

2. Attempt the problem first before using AI

“Desirable difficulties” create durable learning

3. Anthropic Safety Interaction Patterns

Conceptional questioning = test score 65-86%

Full delegation = test scores 24-39%

Unreliable Narration - Mitigations

4. No-AI Day(s)

Compare AI-on performance to AI-off

Similar tasks

Track speed, errors, confidence, actual outcomes

5. Measure what matters

Bugs in PROD, INCs (especially security), rollbacks

NOT: commits, PRs, lines of code

Unreliable Narration - Mitigations

6. Explanation-based code reviews

“Walk me through how this works”

Not just “Does it past CI/CD tests?”

Forces the dev to demonstrate understanding

7. AI-on/AI-off A/B testing at team level as well

Measure: bugs, INCs

Detect if “AI productivity” is eroding quality

Unreliable Narration - Mitigations

- 8. AI is change, so use change management
 - Not: “here’s the tool, now go faster”
 - But: “here’s how cognition shifts & how to detect”
- 9. Anthropic AI Fluency Index Training
 - 11 behaviours that predict effective AI use
 - e.g. questioning outputs leads to better outputs

Unreliable Narration - Mitigations

10. Train your users

ManpowerGroup: 56% received zero AI training

Include all previous mitigations

Show them this presentation :)

That's All For Now

We've covered the most important mechanism.
There are four more in the appendices.

If You Only Remember One Thing

The Touchstone Test is at the heart of this.

Can you explain how the answer was arrived at, without going back to check the chat?

Yes: You learned, AI scaffolded your learning. **Safe.**

No: AI substituted for your learning. **Not safe.**

AI Requires Safety-Critical Thinking

Aviation doesn't ban autopilot.

It mandates minimum unaided practice.

Medicine doesn't ban diagnostic AI.

It mandates clinician benchmarking without AI.

The question isn't "Should we use AI?"

The question is "At which level, for which task, with which safeguards?"

Thank You, and Questions

Questions?

joe.woodhouse@primadonnaconsulting.com

<https://www.linkedin.com/in/joewoodhouseengineer>

<https://github.com/Prima-Donna-Consulting/presentations/blob/current/2026.03.13.You%20Can't%20Outsource%20Judgement.Joe%20Woodhouse.DevOpsNotDead.pdf>

Appendix 0x00

What Didn't Fit in 25m

TBA

[remaining mitigations added here]

[intention is that this will be complete on the day]

Appendix A

Summary of Mitigations

Mitigations: Unreliable Narration

Individual:

- Touchstone Test; conceptual questioning

Departmental:

- Measure outcomes not velocity

- Explanation-based review

Organisational:

- Close 56% training deficit

- Treat as change management

Mitigations: [TBA]

[Will add as each mechanism is added to Ox00]

Appendix B

Which AI Level, and When?

Recap: Three Levels of AI

Level 1: AI Assist (“Help me as I do it”)

Example: Syntax completion

Risk: Low (you retain cognitive load)

Level 2: AI Augment (“Do some of it for me”)

Example: “Write this function”

Risk: HIGHEST (ind), unreliable narration + skill decay

Level 3: Ai Outsource (“Do it, tell me when done”)

Example: Vibe-coding

Risk: HIGHEST (org), huge blast radius

Recap: Three Levels of AI

Level 2 is the most dangerous to individuals
We believe we're learning; we're not

Level 3 is the most dangerous to organisations
Trading security for velocity; cost to remediate

Use of AI Decision Tree

I have a task to complete.

Q1: Is this task security-critical?
(Authentication, authorisation, payments, privacy)

Yes: STOP. Level 0. No AI. <END>

No: <Continue to Q2>

Use of AI Decision Tree

I have a task to complete.

Q2: Do I need to learn/maintain skill in this domain?

Yes: Level 1 only (Assist). The work must be mine.

<END>

No: <Continue to Q3>

Use of AI Decision Tree

I have a task to complete.

Q3: Is this high stakes if it goes wrong?
(Production, customer-facing, compliance)

Yes: Level 1-2 max (Assist/Augment).
Mandatory human review. <END>

No: <Continue to Q4>

Use of AI Decision Tree

I have a task to complete.

Q4: Can I effectively review/validate the output?
(Do I have the expertise to catch errors?)

Yes: <Continue to Q5>

No: **STOP. Don't use AI at all.**
We can't validate what we don't understand.
<END>

Use of AI Decision Tree

I have a task to complete.

Q5: Does the task require Production access?
Or destructive actions?

Yes: Level 1-2 max (Assist/Augment), no agents.
Mandatory approval gates. <END>

No: Level 3 acceptable in sandbox/non-Prod.

Examples

Task	Recommended Level	Rationale
Learning something new	1 (Assist)	Can't outsource learning
Debugging	1-2 (Assist/Augment)	Watch for skill decay; use the Touchstone Test
Generating boilerplate	2 (Augment)	Low value, high repetition = good target for L2
Architecture design	0-1 (No AI/Assist)	High value, must retain expertise
Security code	0 (No AI)	45% vulnerabilities unacceptable
Refactoring	1 (Assist)	High skill, must retain expertise
Generating tests	2 (Augment)	Review for coverage; consider AI vs. AI
Documentation	2 (Augment)	Lower stakes but verify
Code review	0-1 (No AI/Assist)	Human in the loop critical; AI can highlight but not decide
Deploy to PROD	0-2 (No AI/Assist)	Stakes too high for L3 agents
Exploration/prototype	2-3 (Augment/Outsource)	Sandbox only; gain stakeholder agreement to not deploy

Appendix C

References, full

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Appendix D

References, by topic

References by topic

Cognitive Mechanisms (M1, M2, M4)

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References by topic

Security & Code Quality (M3)

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References by topic

Agent Catastrophe (M5)

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Appendix E

About This Presentation

Constraints & Approaches

Time constraint: only 25m for main presentation.

Deep dive on one idea. Rest into appendix.

Density choice: high information density per slide.

Expert audience; must be its own reference.

BLUF approach: conclusion first, then evidence.

Appendix structure

Appendix 0x00 for unstable/overflow.

Appendices A-G stable if overflow re-merged.

Hex numbering visually distinct.

Constraints & Approaches

Accessibility commitment

- Text-only for screen readers

- High-contrast design, 7:1, WCAG AA compliant

- Dark mode most suitable for cinema projection

Open source commitment

- No paywalling or gatekeeping for full deck.

- Publicly available on GitHub.

- Licensed CC BY-NC-SA 4.0

- Dual-licensed for commercial use.

How To Use This Deck

For conference attendees: you saw slides 1-27

Appendix 0x00 has what we didn't have time for

For later readers (GitHub, LinkedIn)

Appendix 0x00 migrating into main

For decision-making on AI use

Start with Appendix A for Mitigation Summary

Use Appendix B for day-to-day decision

Measure what matters!

Appendix F

Use of AI

Use of AI in this presentation

This presentation about AI harms was created with AI assistance. Irony acknowledged.

Here's exactly how AI was used

Level 1 (Assist): Used Extensively

- Literature search

- Citation formatting

- Syntax checking and grammatical review

- Graphical design recommendations & structure

Use of AI in this presentation

Level 2 (Augment): Used Sparingly and Selectively
Outline generation (then heavily edited by human)
Evidence synthesis across all research
“What have I missed here?”
“What questions should I be asking that I didn’t?”

Level 3 (Outsourcing): Never Used
Every slide rewritten and typed by hand
(Except References which were copied/pasted)

Use of AI in this presentation

Why This Matters

Practice what I preach; appropriate use of AI
Retain cognitive load on synthesis and argument
Show that transparency is necessary and possible

Touchstone Test

Can I explain everything with rereading the chat?
YES. Original ideas, reviews, and edits are mine.
AI covered the field of research.

Appendix G

About the author

Joe Woodhouse

Principal Engineer | Non-Executive Director

Background in databases & consulting

Now Infrastructure/Systems/Performance Engineer

“From the server room to the board room”

joe.woodhouse@primadonnaconsulting.com

<https://www.linkedin.com/in/joewoodhouseengineer/>